

RSDC: Regime-Switching Correlation Models in R

by David Ardia, Benjamin Seguin, and Roosevelt Ymele Nguemo

Abstract The **RSDC** package implements regime-switching correlation models in R, extending the Pelletier (2006) framework with time-varying transition probabilities (TVTP) driven by exogenous covariates and with an arbitrary number of regimes through a multinomial-logit (softmax) parameterization. Estimation combines a global search in a reparameterized space in which every candidate is a valid correlation matrix with local refinement and a multi-start replication diagnostic, making the maximum-likelihood estimator usable in higher cross-sectional dimensions. We present the methodology, the package design, a Monte Carlo validation, and a strictly out-of-sample study of five Fama-French industry portfolios in which transitions respond to two nearly orthogonal covariates, the Media Climate Change Concerns index and the VIX, and feed regime-based minimum-variance and maximum-diversification allocations. All results are reproducible from data shipped with the package.

1 Introduction

Modeling time-varying correlations between asset returns is central to portfolio construction and risk management. Smooth-transition approaches such as the DCC model of Engle (2002) let correlations evolve continuously. An alternative view is that correlations spend long spells near regime-specific levels and move abruptly when the market changes state. Regime-switching dynamic correlation (RSDC) models (Pelletier, 2006) formalize this view: the covariance matrix is decomposed into univariate volatilities and a correlation matrix R_S , driven by a latent Markov state S_t , so the correlation matrix is constant within a regime and shifts discretely between regimes.

The **RSDC** package implements this model class and extends it in two directions. First, transition probabilities may be *time-varying* (TVTP): they respond to any observable indicator supplied by the user, a volatility index, a sentiment or news-attention series, a macroeconomic factor, following the state-dependent Markov-switching tradition of Filardo (1994), Diebold et al. (1994) and Gray (1996). Second, all specifications support an arbitrary number of regimes N , with a logistic link for $N = 2$ and a multinomial-logit (softmax) link for $N \geq 3$. In the empirical illustration the covariates are the Media Climate Change Concerns index (Ardia et al., 2023) and the VIX, but nothing in the package is specific to climate applications.

Three model variants are provided: a constant-correlation baseline ("const"), the fixed-transition Markov-switching model of Pelletier (2006) ("noX"), and the covariate-driven TVTP model ("tvtp"). Estimation is by maximum likelihood through the Hamilton filter (Hamilton, 1989). A practical obstacle in this model class is that the likelihood must be maximized over correlation matrices: a box of pairwise correlations almost never contains a positive-definite point once K grows, which starves any box-constrained global optimizer. Since version 1.7-0 the package removes this obstacle by searching in the canonical partial-correlation parameterization of Joe (2006), in which *every* point of the search box is a valid correlation matrix; the full estimation strategy is described in the Methodology section.

Several CRAN packages address neighbouring problems; Table 1 summarizes the comparison. **rmgarch** (Galanos, 2024) implements the DCC family, in which correlations evolve smoothly and transitions cannot be tied to covariates. **MSGARCH** (Ardia et al., 2019) provides Markov-switching GARCH for univariate volatility. General hidden-Markov toolkits such as **depmixS4** (Visser and Speekenbrink, 2010) and **MSwM** can represent switching in generic parameters but offer neither correlation-matrix regimes with guaranteed positive definiteness nor portfolio tooling. **RSDC** fills this gap: regime-switching of the correlation matrix itself, covariate-driven transitions, and a direct pipeline from correlation forecasts to minimum-variance and maximum-diversification portfolios.

Table 1: Related CRAN packages. ‘depmixS4’ can represent switching in generic distribution parameters but has no notion of a positive-definite correlation-matrix regime.

Package	Model class	Corr. regimes	Covar. transitions	Portfolio tools
RSDC	RS correlation	yes	yes	yes
rngarch	DCC (smooth)	no	no	no
MSGARCH	MS-GARCH	no	no	no
depmixS4	hidden Markov	generic	no	no
MSwM	MS regression	no	no	no

Throughout, the input data are *standardized residuals*: returns filtered by univariate volatility models (e.g., GARCH) and scaled to unit variance, so that the regime dynamics identified by the package pertain to correlations only. The remainder of the paper presents, in turn, the methodology, the package design, a simulation-based validation, and the empirical illustration.

2 Methodology

2.1 Regime-switching correlation model

Let U_t denote the K -dimensional vector of standardized residuals at time t and $S_t \in \{1, \dots, N\}$ a latent regime indicator. Conditional on the regime, U_t is multivariate normal with a regime-specific correlation matrix:

$$U_t \mid (S_t = j) \sim \mathcal{N}(0, R_j),$$

where each R_j is a $K \times K$ correlation matrix with $K(K-1)/2$ free parameters $\rho^{(j)}$. With $N = 1$ the model reduces to the constant-correlation baseline ("const"). For identifiability, the estimators order regimes by ascending mean correlation, so regime 1 is always the lowest-correlation state; this labeling convention is used in all outputs.

2.2 Markov switching dynamics

The regime follows a first-order Markov chain. We write P for a constant transition matrix and $\pi_{ij}(t)$ for the operative transition probability at time t : the constant entry P_{ij} in the fixed-transition model, or the covariate-dependent value defined below in the TVTP model.

Fixed transitions ("noX")

The transition matrix P , with $P_{ij} = \Pr(S_t = j \mid S_{t-1} = i)$, is constant. For $N = 2$,

$$P = \begin{pmatrix} p_{11} & 1 - p_{11} \\ 1 - p_{22} & p_{22} \end{pmatrix},$$

with p_{11}, p_{22} the regime persistences. For $N \geq 3$, row i is parameterized by its first $N-1$ probabilities, the last entry following by complement; each free probability is bounded in $[0.01, 0.99]$.

Time-varying transitions ("tvtp")

With covariates $x_t \in \mathbb{R}^p$ (including an intercept column if desired, the package does not add one automatically), the transition matrix varies over time. For $N = 2$ the *stay* probabilities follow a logistic link,

$$\pi_{ii}(t) = \frac{\exp(x_t^\top \beta_i)}{1 + \exp(x_t^\top \beta_i)}, \quad i = 1, 2,$$

with the switching probability obtained by complement, $\pi_{ij}(t) = 1 - \pi_{ii}(t)$ for $j \neq i$. Here β_i is a length- p coefficient vector; when x_i contains an intercept and a single indicator, a large positive slope in β_1 means that regime 1 becomes more persistent when the indicator is high.

For $N \geq 3$ a multinomial-logit (softmax) link is used: for each current state i ,

$$\pi_{ij}(t) = \frac{\exp(x_i^\top \beta_{ij})}{\sum_{k=1}^N \exp(x_i^\top \beta_{ik})}, \quad j = 1, \dots, N,$$

with the N -th logit fixed at zero as the reference category, leaving $N - 1$ free coefficient vectors β_{ij} per row. The $N = 2$ logistic link is this softmax specialized to two states with state i itself as the modeled category, so the two displays use one convention. The coefficient index carries the difference: one vector β_i per row when $N = 2$, $N - 1$ vectors β_{ij} per row when $N \geq 3$.

Model variants and parameter count

The three variants differ only in their transition block. The constant-correlation model (method = "const") is the $N = 1$ special case: a single matrix R_1 estimated over the whole sample, whose log-likelihood admits a closed form; it is the natural baseline. The fixed-transition model (method = "noX"), following [Pelletier \(2006\)](#), captures regime shifts without exogenous covariates. The TVTP model (method = "tvtp") lets regime durations depend explicitly on observed covariates such as financial-stress indicators, volatility indices, or news-attention series.

The number of estimated parameters is:

- **TVTP:** transition coefficients, $N \times p$ for $N = 2$ (logistic) or $N(N - 1) \times p$ for $N \geq 3$ (softmax), plus N correlation matrices, i.e. $NK(K - 1)/2$ correlations in total;
- **Fixed transitions (noX):** the $N(N - 1)$ free entries of P plus the $NK(K - 1)/2$ correlations;
- **Constant correlation (const):** a single correlation matrix, $K(K - 1)/2$ parameters;

where p is the number of columns of x_t , including the intercept if one is supplied. Means and variances are not estimated: inputs are assumed standardized, and the fitted objects report placeholder regime means (0) and volatilities (1) that do not enter the likelihood. Model selection can be guided by information criteria such as the AIC and BIC.

2.3 Likelihood inference via the Hamilton filter

Parameters are estimated by maximum likelihood, with the likelihood evaluated by the Hamilton filter ([Hamilton, 1989](#)). The filter proceeds as follows.

Prediction step. Given the filtered probability from the previous time step, $\xi_{i,t-1} = \Pr(S_{t-1} = i \mid \Omega_{t-1})$, where Ω_{t-1} is the information available up to time $t - 1$, the predictive probability of being in state j at time t is

$$\Pr(S_t = j \mid \Omega_{t-1}) = \sum_{i=1}^N \xi_{i,t-1} \pi_{ij}(t),$$

with $\pi_{ij}(t)$ the operative transition probability defined above.

Update step. We observe U_t and compute its likelihood under each regime,

$$f(U_t \mid S_t = j) = (2\pi)^{-K/2} |R_j|^{-1/2} \exp\left\{-\frac{1}{2} U_t^\top R_j^{-1} U_t\right\},$$

and Bayes' rule updates the filtered probabilities:

$$\xi_{j,t} = \Pr(S_t = j \mid \Omega_t) = \frac{\Pr(S_t = j \mid \Omega_{t-1}) f(U_t \mid S_t = j)}{\sum_{m=1}^N \Pr(S_t = m \mid \Omega_{t-1}) f(U_t \mid S_t = m)}.$$

Likelihood. The log-likelihood accumulates the prediction-error decomposition,

$$\sum_{t=1}^T \log f(U_t \mid \Omega_{t-1}), \quad f(U_t \mid \Omega_{t-1}) = \sum_{j=1}^N \Pr(S_t = j \mid \Omega_{t-1}) f(U_t \mid S_t = j).$$

Initialization and smoothing. At $t = 1$ the filter uses a uniform prior $\xi_{i,0} = 1/N$; an optional warm-start state (`xi_init`) is used internally by the two-pass out-of-sample forecast. After filtering, a backward recursion delivers the smoothed probabilities $\Pr(S_t = j \mid \Omega_T)$. The filter's inner loop is implemented in C++ (via [RcppArmadillo](#)), with the pure-R implementation retained as a tested reference.

2.4 A feasible-by-construction global search

The likelihood surface of a switching model is multimodal, and its natural parameter space is hostile to global search: a correlation matrix must be positive definite, which is a *joint* constraint on the $K(K-1)/2$ pairwise correlations that no per-coordinate box can express. The probability that a uniform draw from the box is a valid matrix collapses with K , near zero already for $K \geq 7$, so a naive global optimizer starts from an almost entirely infeasible population. Since version 1.7-0 the package therefore searches in a reparameterized space in which feasibility holds *by construction*, with three ingredients.

Partial correlations. The global stage explores the canonical partial correlations of a C-vine ([Joe, 2006](#)): a vector $z \in (-1, 1)^{K(K-1)/2}$ whose map to correlation matrices is a *bijection* with the set of positive-definite correlation matrices, every point of the box is a valid model at any K . The first $K-1$ coordinates are plain correlations with the first variable, deeper coordinates are partial correlations, so the parameterization remains interpretable; it is the same transform used by Stan for its correlation matrix type.

Bounded-softmax transition head. For "noX" with $N \geq 3$ the free transition probabilities of each row face a second joint constraint (a row must sum to at most one). The global stage searches per-row softmax logits bounded at ± 4 instead: every point of the box now yields a valid row, and the bound, the historical $[0.01, 0.99]$ probability box mapped to the log-odds scale, keeps every transition probability away from zero, so no regime can degenerate into an unreachable state during the search. The $N = 2$ head, which faces no joint constraint, is unchanged.

Search, refinement, and replication. Each search runs differential evolution ([Mullen et al., 2011](#); [Ardia et al., 2011](#)) in the reparameterized space, refines the best mutually distant candidates of the final population by L-BFGS-B in the natural space, and returns the best refined point. Because a single stochastic search can still land in a suboptimal basin of a multimodal surface, `control = list(n_starts = k)` runs k independent searches from derived seeds and keeps the best fit; the per-start log-likelihoods are stored on the fit, and a warning is issued when the maximum is not replicated by at least two searches. Data-driven warm starts (`rsdc_starts()`) remain available as a deterministic complement: when both routes agree, the optimum is cross-certified. All searches are reproducible: the seed is exposed via `control$seed` (default 123) and results are invariant to the number of cores.

Model selection across fitted specifications proceeds by information criteria; because the three variants are nested (set covariate slopes to zero to obtain "noX"; collapse regimes to obtain "const"), likelihood-ratio tests are available with base R alone, as the empirical illustration shows. A free by-product of nesting is a consistency check: a nested model can never attain a higher maximized likelihood than the model that contains it, so any violation reveals an under-optimized fit.

3 Package design and usage

The user-facing API is small. Table 2 lists the exported functions; everything else (S3 methods, tidiers) follows R conventions.

Table 2: Exported functions of RSDC.

Function	Purpose
<code>rsdc_estimate()</code>	fit const, noX or tvtp by maximum likelihood
<code>rsdc_starts()</code>	data-driven warm starts (multi-start)
<code>rsdc_likelihood()</code>	likelihood evaluation
<code>rsdc_hamilton()</code>	Hamilton filter and smoother
<code>rsdc_forecast()</code>	correlation and covariance paths
<code>rsdc_forecast_ahead()</code>	multi-step-ahead forecasts
<code>rsdc_corr_bands()</code>	uncertainty bands on the correlation path
<code>rsdc_viterbi()</code>	most likely (MAP) regime path
<code>rsdc_bootstrap()</code>	parametric bootstrap (SEs, CIs)
<code>rsdc_simulate()</code>	simulate from a fitted or specified model
<code>rsdc_minvar()</code> , <code>rsdc_maxdiv()</code>	long-only MV and MD portfolio weights

`rsdc_estimate()` takes `method`, `residuals`, `N`, `X`, `out_of_sample` and a control list, and returns an "rsdc_fit" object storing, among others, correlations (the $N \times K(K-1)/2$ regime correlations), `transition_matrix` (for "tvtp", evaluated at the in-sample covariate means, a point summary, not the path), `beta`, `log_likelihood` (and `log_likelihood_oos` when `out_of_sample = TRUE`), and filtered/smoothed probabilities. Standard generics apply: `print`, `summary`, `coef`, `logLik`, `AIC`, `BIC`, `nobs`, `vcov`, `confint`, `predict`, `simulate`, `plot`, plus broom-style `tidy()`/`glance()`/`augment()` and a `ggplot2::autoplot()` method. The control list exposes the estimation protocol: `seed` (default 123), `n_starts`, `cores`, `itermax/NP/steptol` (global-stage budget), `maxit` (refinement), `compute_se`, and `start` (a warm start, which skips the global search).

As a compact walkthrough, and a first look at the data machinery used in the empirical illustration, we fit the two-regime TVTP model to the green/brown portfolio pair shipped with the package (Ardia et al., 2023), with the Media Climate Change Concerns (MCCC) index as the transition covariate. Returns are first filtered by univariate GJR-GARCH(1,1) models (`rugarch`); to keep the walkthrough strictly out-of-sample we estimate every model on the first 70% of the sample (`out_of_sample = TRUE`) and reserve the remainder for evaluation, scaling the covariate by its training-window moments.

```
library(RSDC)
library(rugarch)

data("greenbrown"); data("mccc")
stopifnot(identical(greenbrown$DATE, mccc$DATE))
gb_ret <- as.matrix(greenbrown[, c("return_green", "return_brown")]) * 100
n_gb   <- nrow(gb_ret)
cut_gb <- round(0.7 * n_gb)                    # the package's 70/30 split

# GJR-GARCH(1,1), zero mean, fitted on the training window only; the fitted
# parameters then filter the full series (no test-window information enters).
garch_filter <- function(ret, cut) {
  spec <- ugarchspec(
    variance.model = list(model = "gjrGARCH", garchOrder = c(1, 1)),
    mean.model     = list(armaOrder = c(0, 0), include.mean = FALSE))
  K <- ncol(ret)
  z <- s <- matrix(NA_real_, nrow(ret), K,
    dimnames = list(NULL, colnames(ret)))
  for (i in seq_len(K)) {
    fit <- ugarchfit(spec, ret[, i], solver = "hybrid",
      out.sample = nrow(ret) - cut)
    spec_i <- spec; setfixed(spec_i) <- as.list(coef(fit))
    filt <- ugarchfilter(spec_i, ret[, i])
    z[, i] <- as.numeric(residuals(filt)) / as.numeric(sigma(filt))
    s[, i] <- as.numeric(sigma(filt))
  }
}
```

```

    }
    for (i in seq_len(K))                # train-window scale
      z[, i] <- z[, i] / sd(z[seq_len(cut), i])
    list(z = z, s = s)
  }
gb <- garch_filter(gb_ret, cut_gb)
mccc_tr <- mccc$mccc
mccc_std <- (mccc_tr - mean(mccc_tr[1:cut_gb])) /
  sd(mccc_tr[1:cut_gb])
X_gb <- cbind(intercept = 1, MCCC = mccc_std)

fit_gb <- rsdc_estimate("tvtp", residuals = gb$z, N = 2, X = X_gb,
  out_of_sample = TRUE,
  control = list(n_starts = 4, cores = 4))

fit_gb

#> RSDC fit: method = "tvtp", N = 2 regime(s), K = 2 series, T = 1586
#> Call: rsdc_estimate(method = "tvtp", residuals = gb$z, N = 2, X = X_gb,
#>   out_of_sample = TRUE, control = list(n_starts = 4, cores = 4))
#> logLik = -4172.011   npar = 6   AIC = 8356.02   BIC = 8388.24
#>
#> Regime correlations (rows = regimes, ascending mean):
#>      [,1]
#> [1,] 0.2026
#> [2,] 0.7474
#>
#> Transition matrix (evaluated at the in-sample covariate means):
#>      [,1] [,2]
#> [1,] 0.8500 0.1500
#> [2,] 0.0899 0.9101
#>
#> Optimiser convergence code: 52 (0 = converged)

```

The printed fit reports the regime correlations (ordered low to high), the transition summary, and both the training log-likelihood and the out-of-sample predictive log-likelihood. `summary(fit_gb)` adds standard errors from the observed information, delta-method standard errors for the implied transition probabilities, and regime diagnostics (expected durations, ergodic distribution); `confint()` provides Wald intervals. Regimes beyond three are supported for "noX" and "tvtp" ($N = 4$ and higher), with the same softmax head.

4 Validation

Two kinds of evidence support the implementation. First, an analytic anchor: with `method = "const"` the maximum-likelihood correlation matrix is the sample correlation of the standardized residuals, and the package reproduces it to numerical precision (this identity is part of the test suite). Second, a full Monte Carlo study ships with the package: the script `inst/simulation/monte_carlo_study.R` regenerates it end to end, and its report `montecarlo_validation.pdf` is installed alongside and available online at https://github.com/ArdiaD/RSDC/blob/master/inst/simulation/montecarlo_validation.pdf. The study covers five specifications, constant, noX and tvtp at $N \in \{2, 3\}$, simulated and re-estimated with $M = 100$ replications per cell, $T \in \{500, 1000, 2000\}$ and, for the three-regime fixed-transition case, $K \in \{2, 3, 5\}$.

Three findings matter for users. Bias and RMSE shrink with T in every cell, at the $\sqrt{T}$ rate where the surface is regular, and every cell converges in all 100 replications, including $K = 5$ (33 parameters), a dimension at which the pre-1.7-0 natural-space search produced no usable replication at all. Median RMSE *falls* as K grows (each added series is one more observation on the same latent regime path), which turns the conjecture that regime-switching correlation

models pay off increasingly with dimension into a measured result. Finally, in the three-regime TVTP cells a fraction of replications reach the ± 10 coefficient bound at small T , the classical separation problem of multinomial-logit transitions when some transitions are observed only a handful of times, and that fraction falls from 75% to 36% of replications as T doubles; the study's usable counts report it transparently.

5 Empirical illustration

The main application uses the `ff5ind` data set shipped with the package: daily value-weighted returns (percent) of five Fama-French industry portfolios, Manufacturing, Energy, High Tech, Health, Utilities (Fama and French, 1997), together with the MCCC index (monthly Aggregate, forward-filled across trading days) and the CBOE VIX, aligned on $T = 5155$ U.S. trading days from 2005-01-03 to 2025-06-30. The two candidate transition covariates are nearly orthogonal in this window, which lets the data separate two mechanisms: market fear (VIX) and climate attention (MCCC).

Everything below is out-of-sample in the strict sense: the GARCH pre-filter, the covariate scaling, and all regime-model parameters are estimated on the first 70% of the sample ($T_{\text{train}} = 3608$ days, 2005-2019); the remaining 30% ($T_{\text{test}} = 1547$ days, 2019-2025, spanning the COVID crash and the 2022 tightening cycle) is used only for evaluation, with out-of-sample correlations weighted by *filtered* (not smoothed) probabilities so no future information enters.

5.1 Data and preprocessing

```
data("ff5ind")
industries <- c("Manuf", "Enrgy", "HiTec", "Hlth", "Utils")
ret <- as.matrix(ff5ind[, industries])
dates <- ff5ind$DATE
n <- nrow(ret)
cut <- round(0.7 * n) # matches the estimators' split
c(train_end = format(dates[cut]), test_start = format(dates[cut + 1]))

#> train_end test_start
#> "2019-05-03" "2019-05-06"

ff5 <- garch_filter(ret, cut) # train-window GARCH parameters
z <- ff5$z; s <- ff5$s

std_train <- function(x) (x - mean(x[1:cut])) / sd(x[1:cut])
X_mccc <- cbind(intercept = 1, MCCC = std_train(ff5ind$mccc))
X_vix <- cbind(intercept = 1, logVIX = std_train(log(ff5ind$vix)))
X_both <- cbind(X_mccc, logVIX = X_vix[, "logVIX"])
round(cor(X_mccc[, "MCCC"], X_vix[, "logVIX"]), 3) # near-orthogonal covariates

#> [1] 0.003
```

5.2 Estimation and model selection

We estimate nine specifications with the same call, all with `out_of_sample = TRUE` and the multi-start protocol; per-model wall-clock times are collected for the timing table below.

```
ctrl <- list(n_starts = 4, cores = 14, compute_se = FALSE)
fit1 <- function(method, N, X = NULL) {
  t0 <- Sys.time()
  f <- rsdc_estimate(method, residuals = z, N = N, X = X,
                    out_of_sample = TRUE, control = ctrl)
  attr(f, "minutes") <- as.numeric(Sys.time() - t0, units = "mins")
}
```



```

    f
  }
fits <- list(
  "const"      = fit1("const", 1),
  "noX(N=2)"   = fit1("noX", 2),
  "noX(N=3)"   = fit1("noX", 3),
  "tvtp2-MCCC" = fit1("tvtp", 2, X_mccc),
  "tvtp2-VIX"  = fit1("tvtp", 2, X_vix),
  "tvtp2-both" = fit1("tvtp", 2, X_both),
  "tvtp3-MCCC" = fit1("tvtp", 3, X_mccc),
  "tvtp3-VIX"  = fit1("tvtp", 3, X_vix),
  "tvtp3-both" = fit1("tvtp", 3, X_both))

comparison <- data.frame(
  model      = names(fits),
  logLik     = sapply(fits, function(f) as.numeric(logLik(f))),
  npar       = sapply(fits, function(f) f$npars),
  AIC        = sapply(fits, AIC),
  BIC        = sapply(fits, BIC),
  logLikOOS  = sapply(fits, function(f) f$log_likelihood_oos),
  row.names = NULL)
knitr::kable(comparison, digits = 1, caption = sprintf(
  "Model comparison, five Fama-French industries. Training window
  2005-%s (T = %d): log-likelihood, parameters, AIC, BIC. Last column:
  predictive log-likelihood on the out-of-sample window (T = %d), computed with
  training-window parameters and filtered probabilities.",
  format(dates[cut], "%Y"), cut, n - cut))

```

Table 3: Model comparison, five Fama-French industries. Training window 2005-2019 (T = 3608): log-likelihood, parameters, AIC, BIC. Last column: predictive log-likelihood on the out-of-sample window (T = 1547), computed with training-window parameters and filtered probabilities.

model	logLik	npars	AIC	BIC	logLikOOS
const	-19785	10	39590	39652	-10686
noX(N=2)	-19155	22	38355	38491	-10153
noX(N=3)	-19004	36	38080	38302	-10017
tvtp2-MCCC	-19153	24	38355	38503	-10132
tvtp2-VIX	-19108	24	38264	38413	-10183
tvtp2-both	-19104	26	38260	38421	-10116
tvtp3-MCCC	-18992	42	38067	38327	-10272
tvtp3-VIX	-18948	42	37981	38241	-10190
tvtp3-both	-18941	48	37978	38275	-10031

Three regimes dominate two on every criterion, and every switching model dominates the constant baseline. The criteria disagree, however, on the transition specification: the AIC selects the richest model (three regimes, both covariates), the BIC, penalizing harder, prefers the VIX-only variant, and the out-of-sample predictive log-likelihood is best for the fixed-transition noX(N=3) model, with tvtp3-both a close second (a 14 log-point gap over 1547 days). In-sample, the covariates add substantial information about transitions; that edge does not translate into better prediction on this particular hold-out, which opens with the COVID crash, a regime entry unlike anything in the training window. We retain tvtp3-both, the AIC choice, and the specification that lets us study the covariate effects, as the working model, and report the fixed-transition alternative alongside it in the portfolio evaluation. Because the single-covariate models are nested in the two-covariate model, likelihood-ratio tests need only base R:

```

ll <- setNames(comparison$logLik, comparison$model)
np <- setNames(comparison$npars, comparison$model)

```



```

lr <- function(big, small) {
  stat <- 2 * (ll[[big]] - ll[[small]]); df <- np[[big]] - np[[small]]
  c(LR = round(stat, 1), df = df,
    p = signif(pchisq(stat, df, lower.tail = FALSE), 2))
}
rbind("VIX adds to MCCC (N = 3)" = lr("tvt3-both", "tvt3-MCCC"),
      "MCCC adds to VIX (N = 3)" = lr("tvt3-both", "tvt3-VIX"),
      "three regimes vs two"     = lr("tvt3-both", "tvt2-both"))

#>
#> LR df p
#> VIX adds to MCCC (N = 3) 100.7 6 1.8e-19
#> MCCC adds to VIX (N = 3) 14.6 6 2.4e-02
#> three regimes vs two 326.0 22 6.3e-56

```

The two covariates contribute asymmetrically on the training window: the VIX adds overwhelming information given MCCC (LR = 100.7), while MCCC adds significant but modest information given the VIX (LR = 14.6, $p = 0.02$). Their near-orthogonality still matters, whatever MCCC contributes is not market fear in disguise, and the bootstrap below shows that several individual MCCC slopes are precisely estimated. (The three-vs-two-regime statistic is indicative only, as regime-count tests face unidentified parameters under the null; AIC, BIC and the out-of-sample likelihood all point the same way.) As a consistency check, every nested pair in Table 3 satisfies the required likelihood inequality.

5.3 Inference on the retained model

For inference we re-estimate the retained specification on the training window (a warm start from the fit above, so no global search is repeated) and run the parametric bootstrap.

```

fit_ret <- rsdc_estimate("tvt3", residuals = z[1:cut, ], N = 3,
  X = X_both[1:cut, ],
  control = list(start = fits[["tvt3-both"]][par,
    compute_se = FALSE))

set.seed(123)
bs <- rsdc_bootstrap(fit_ret, B = 199, X = X_both[1:cut, ], seed = 123,
  cores = 14)

```

The fitted object also carries observed-information standard errors for every coefficient, delta-method standard errors for the implied transition matrix at the covariate means, and regime diagnostics (expected durations, ergodic distribution); the full `summary()` output is reported in the Appendix.

```

fit_se <- rsdc_estimate("tvt3", residuals = z[1:cut, ], N = 3,
  X = X_both[1:cut, ],
  control = list(start = fit_ret$par))

summary(fit_se)

```

The two inference routes agree: the observed-information standard errors track the bootstrap ones closely (median ratio 1.09), and the two methods classify 11 of the 12 covariate slopes identically at the 5 percent level, the exception being the borderline calm-persistence MCCC slope. We keep the bootstrap intervals as the headline, since several transition probabilities sit near their bounds, where the quadratic approximation behind the observed information is least reliable.

```

ci <- bs$ci
rows <- c(grep("x2", rownames(ci), fixed = TRUE),
  grep("x3", rownames(ci), fixed = TRUE))
tab <- cbind(ci[rows, , drop = FALSE], se = bs$se[rows])
rownames(tab) <- sub("x2", "MCCC", sub("x3", "logVIX", rownames(tab)))

```

```

star <- ifelse(tab[, 1] * tab[, 2] > 0, "*", "")
knitr::kable(cbind(as.data.frame(round(tab, 3)), `` = star), caption = sprintf(
  "Percentile 95 percent bootstrap intervals for the MCCC and log-VIX transition
  slopes of the retained model (B = %d parametric replicates on the
  training window; coefficients are softmax logits with the crisis regime
  as reference; * = interval excludes zero).", bs$B))

```

Table 4: Percentile 95 percent bootstrap intervals for the MCCC and log-VIX transition slopes of the retained model (B = 199 parametric replicates on the training window; coefficients are softmax logits with the crisis regime as reference; * = interval excludes zero).

	2.5%	97.5%	se	
b[g1,c1,MCCC]	0.060	3.155	0.715	*
b[g1,c2,MCCC]	-0.571	2.826	0.756	
b[g2,c1,MCCC]	0.190	1.237	0.278	*
b[g2,c2,MCCC]	0.187	1.253	0.272	*
b[g3,c1,MCCC]	-1.379	2.198	0.906	
b[g3,c2,MCCC]	-2.244	-0.282	0.494	*
b[g1,c1,logVIX]	-3.756	-0.134	0.970	*
b[g1,c2,logVIX]	-2.797	0.897	0.944	
b[g2,c1,logVIX]	-2.111	-0.567	0.383	*
b[g2,c2,logVIX]	-0.676	0.290	0.259	
b[g3,c1,logVIX]	-1.396	1.160	0.591	
b[g3,c2,logVIX]	-3.048	-1.003	0.523	*

Figures 1 and 2 turn the estimated coefficients into pictures; each curve varies one covariate with the other held at its training mean. Against log-VIX, fear empties the calm state and locks in the high-correlation ones: calm persistence falls from about 0.7 in quiet markets to essentially zero at high VIX, while the crisis regime's stay probability climbs from below 0.5 to near certainty. Under stress the system does not merely visit the high-correlation states, it stays there. Against MCCC, attention works mostly through the calm state, whose persistence roughly quadruples across the observed range, so durable low-correlation spells require high climate attention; crisis persistence also rises somewhat. The two covariates thus act on different margins of the transition matrix, consistent with each carrying its own information in the likelihood-ratio tests. We read all of these as conditional associations estimated from one sample, not causal effects.

```

# Stay probability pi_ii(x) from the softmax rows of the estimated beta
# (row i: two destination blocks of (intercept, MCCC, logVIX); ref = crisis)
stay_prob <- function(beta, i, x_mccc, x_vix) {
  x <- c(1, x_mccc, x_vix)
  l1 <- sum(x * beta[i, 1:3]); l2 <- sum(x * beta[i, 4:6])
  e <- exp(c(l1, l2, 0)) - max(l1, l2, 0)
  (e / sum(e))[min(i, 3)]
}
persistence_fig <- function(vary = c("vix", "mccc")) {
  vary <- match.arg(vary)
  obs <- if (vary == "vix") X_vix[1:cut, 2] else X_mccc[1:cut, 2]
  grid <- seq(quantile(obs, 0.005), quantile(obs, 0.995), length.out = 200)
  B <- fit_ret$beta
  d <- do.call(rbind, lapply(1:3, function(i) data.frame(
    x = grid, regime = c("Calm", "Normal", "Crisis")[i],
    p = vapply(grid, function(g)
      if (vary == "vix") stay_prob(B, i, 0, g) else stay_prob(B, i, g, 0),
      0)))
  d$regime <- factor(d$regime, levels = c("Calm", "Normal", "Crisis"))
  lab <- d[ave(d$x, d$regime, FUN = max) == d$x, ]
  vj <- c(Calm = 0.5, Normal = 1.6, Crisis = -0.7)

```

```

lab$vjjust <- vj[as.character(lab$regime)]
xlab_txt <- if (vary == "vix") "log VIX (standardized)" else
  "MCCC (standardized)"
ggplot(d, aes(x, p, colour = regime)) +
  geom_line(linewidth = 0.9) +
  geom_text(data = lab, aes(label = regime, vjust = vjjust),
    hjust = -0.15, fontface = "bold", size = 3.4,
    show.legend = FALSE) +
  geom_rug(data = data.frame(x = obs), aes(x = x), inherit.aes = FALSE,
    sides = "b", alpha = 0.15, length = unit(0.03, "npc")) +
  scale_colour_manual(values = c("#2166ac", "#e8a33d", "#d1541e"),
    guide = "none") +
  scale_x_continuous(expand = expansion(mult = c(0.02, 0.12))) +
  labs(y = "P(stay in regime)", x = xlab_txt) +
  thm + theme(axis.title.x = element_text())
}

persistence_fig("vix")

```

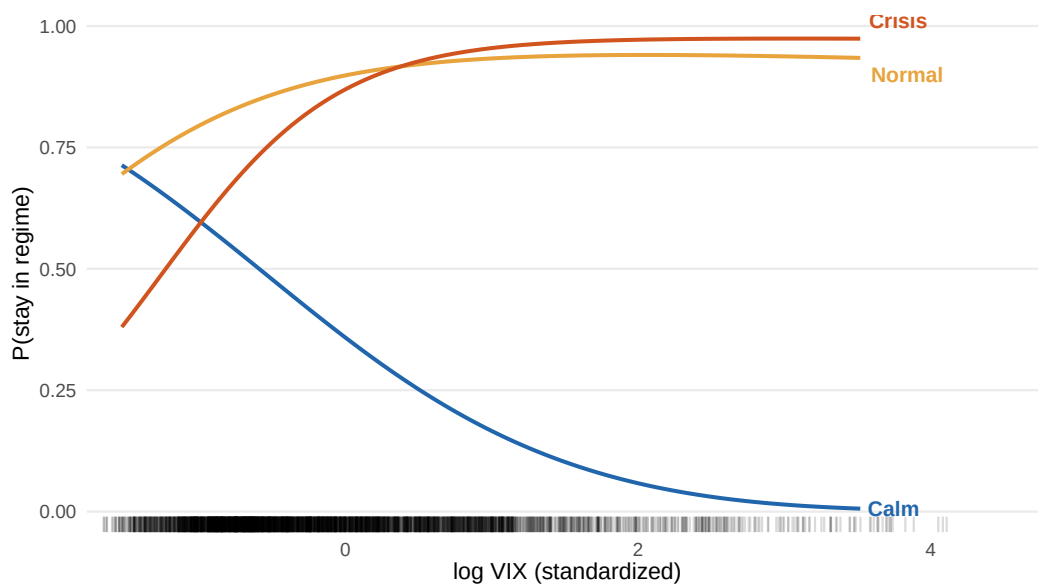


Figure 1: Regime persistence as a function of log-VIX, MCCC held at its training mean; TVTP(N = 3), training-window estimates. Ticks along the x-axis: observed log-VIX distribution.

```

persistence_fig("mccc")

```

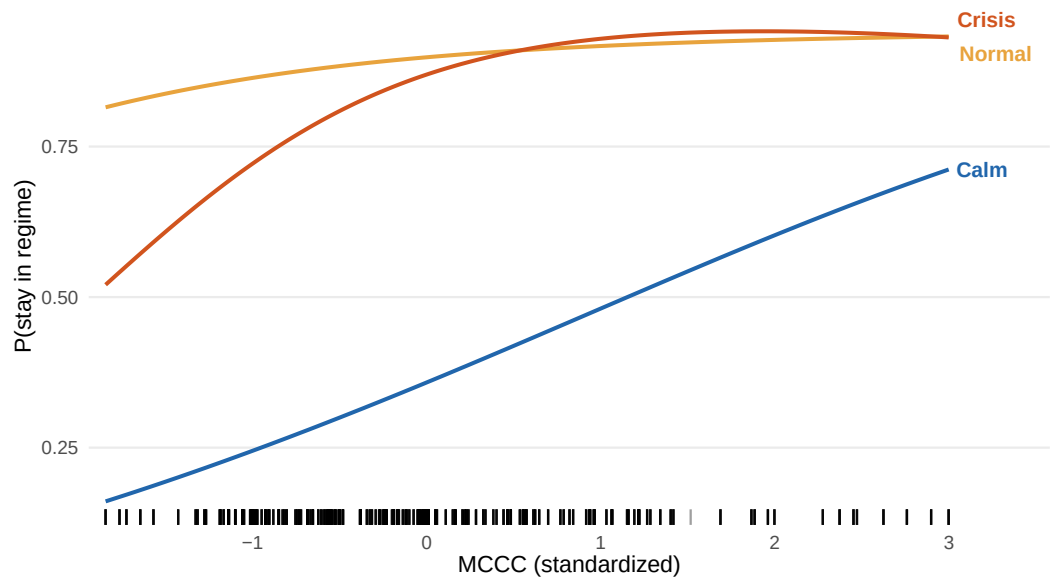


Figure 2: Regime persistence as a function of MCCC, log-VIX held at its training mean; TVTP(N = 3), training-window estimates. Ticks along the x-axis: observed MCCC distribution.

Parameter uncertainty can also be propagated into the correlation path itself. Figure 3 shows the 95 percent band produced by `rsdc_corr_bands()`: parameter vectors are drawn from the estimated sampling distribution of the training-window fit, the filter is rerun for each draw over the full sample, and pointwise quantiles are formed; the out-of-sample segment is shown for one economically interpretable pair, old-economy Energy against new-economy HiTec. The band is tight inside persistent regimes and widens sharply around regime switches, where classification uncertainty dominates parameter uncertainty.

```
bands <- rsdc_corr_bands(fit_se, residuals = z, X = X_both,
                        B = 200, seed = 123)

pair_names <- apply(utils::combn(industries, 2), 2, paste, collapse = "-")
b <- bands[[which(pair_names == "Enrgy-HiTec")]]
keep_oos <- (cut + 1):n
db <- data.frame(date = dates[keep_oos], fit = b[keep_oos, "fit"],
                lo = b[keep_oos, "lower"], hi = b[keep_oos, "upper"])
ggplot(db, aes(date, fit)) +
  geom_ribbon(aes(ymin = lo, ymax = hi), fill = "#1f5fa8", alpha = 0.20) +
  geom_line(colour = "#1f5fa8", linewidth = 0.35) +
  labs(y = "Correlation (Enrgy-HiTec)") + thm
```

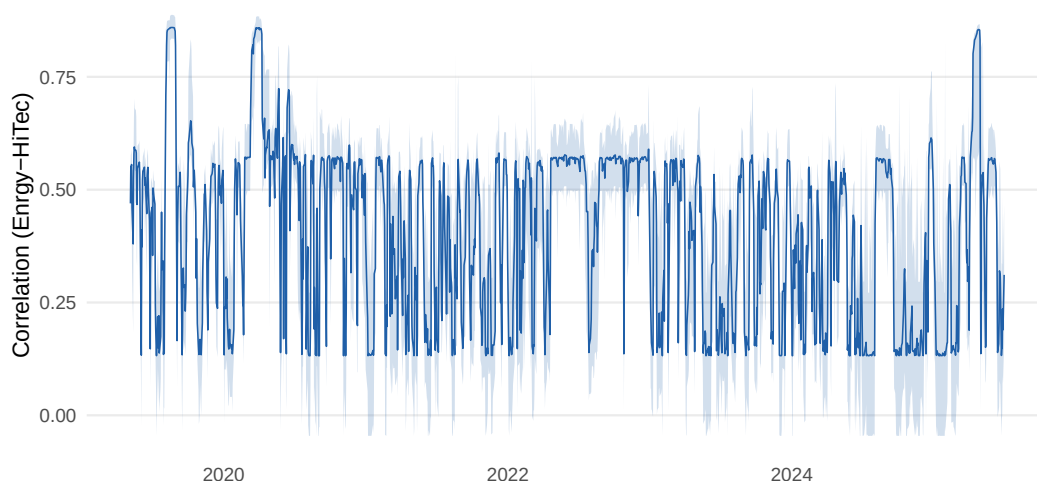


Figure 3: Predicted Energy-HiTec correlation with a 95 percent parameter-uncertainty band ($B = 200$ draws from the training-window sampling distribution), out-of-sample segment.

5.4 Regimes and the out-of-sample correlation path

```
fc <- rsdc_forecast("tvtp", N = 3, residuals = z, X = X_both,
  final_params = fits[["tvtp3-both"]], sigma_matrix = s,
  value_cols = industries, out_of_sample = TRUE)
oos_dates <- dates[(cut + 1):n]
mean_corr_oos <- rowMeans(fc$predicted_correlations)
```

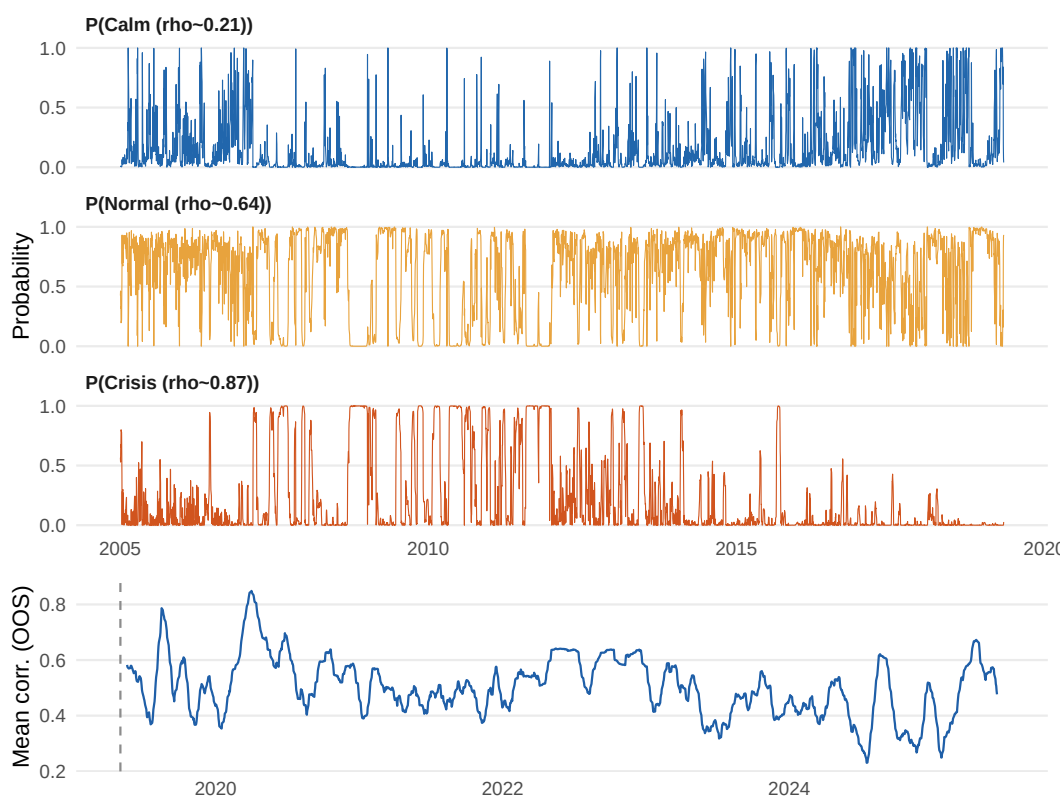


Figure 4: Smoothed regime probabilities of the retained model on the training window (top three panels; regimes ordered by mean correlation) and the mean predicted correlation on the out-of-sample window (bottom; filtered weights, 21-day moving average). The dashed line marks the train/test split.

On the training window the three regimes read as calm, normal and crisis states (mean correlations 0.21 / 0.64 / 0.87), with the 2008-2009 and 2010-2012 episodes classified as crisis spells. Out of sample, the filtered mean-correlation path rises sharply in March 2020 and again through 2022, periods the model has never seen, before easing over 2023-2025.

5.5 Portfolio evaluation on the out-of-sample window

The out-of-sample covariances $\Sigma_t = D_t R_t D_t$ (training-window GARCH parameters around filtered correlations) feed the long-only minimum-variance and maximum-diversification optimizers, evaluated with `lag = TRUE`: weights formed at $t - 1$ earn the period- t return, and since parameters, covariate scaling and volatility filters all come from the training window, the backtest is look-ahead-free in the strict sense.

```
ret_oos <- ret[(cut + 1):n, ]
lagged_ret <- function(W) rowSums(rbind(NA, W[-nrow(W), ] * ret_oos)
pf_pair <- function(fcst) {
  mv <- rsdc_minvar(sigma_matrix = fcst$sigma_matrix, value_cols = industries,
                    predicted_corr = fcst$predicted_correlations, y = ret_oos,
                    long_only = TRUE, lag = TRUE)
  md <- rsdc_maxdiv(sigma_matrix = fcst$sigma_matrix, value_cols = industries,
                    predicted_corr = fcst$predicted_correlations, y = ret_oos,
                    long_only = TRUE, lag = TRUE)
  list(mv = lagged_ret(mv$weights), md = lagged_ret(md$weights))
}
p_tvtp <- pf_pair(fc)
fc_noX <- rsdc_forecast("noX", N = 3, residuals = z,
                       final_params = fits[["noX(N=3)"]], sigma_matrix = s,
                       value_cols = industries, out_of_sample = TRUE)
p_noX <- pf_pair(fc_noX)
pf <- data.frame(`minvar (tvtp3-both)` = p_tvtp$mv,
                 `minvar (noX3)` = p_noX$mv,
                 `maxdiv (tvtp3-both)` = p_tvtp$md,
                 equal = rowMeans(ret_oos), check.names = FALSE)[-1, ]

pf_stats <- t(sapply(pf, function(r) {
  vol <- sd(r) * sqrt(252); ret_a <- mean(r) * 252
  cum <- cumprod(1 + r / 100)
  mdd <- -100 * max(1 - cum / cummax(cum))
  c(`vol (%)` = vol, `return (%)` = ret_a,
    `Sharpe` = ret_a / vol, `max DD (%)` = mdd)
})))
knitr::kable(round(pf_stats, 3), caption = sprintf(
  "Out-of-sample portfolio evaluation (%s to %s, %d days): annualized volatility,
  return and Sharpe ratio of the long-only strategies, minimum variance
  under the retained TVTP model and under the fixed-transition noX(N=3)
  alternative, maximum diversification under the retained model, against
  the equal-weight benchmark. Weights at t-1 earn the t return; all
  parameters are training-window estimates.",
  format(oos_dates[2]), format(max(oos_dates)), nrow(pf)))
```

Table 5: Out-of-sample portfolio evaluation (2019-05-07 to 2025-06-30, 1546 days): annualized volatility, return and Sharpe ratio of the long-only strategies, minimum variance under the retained TVTP model and under the fixed-transition noX(N=3) alternative, maximum diversification under the retained model, against the equal-weight benchmark. Weights at t-1 earn the t return; all parameters are training-window estimates.

	vol (%)	return (%)	Sharpe	max DD (%)
minvar (tvtp3-both)	17.33	11.36	0.656	-31.92
minvar (noX3)	17.38	11.26	0.648	-31.89
maxdiv (tvtp3-both)	19.33	13.26	0.686	-37.36
equal	20.50	14.12	0.689	-38.67

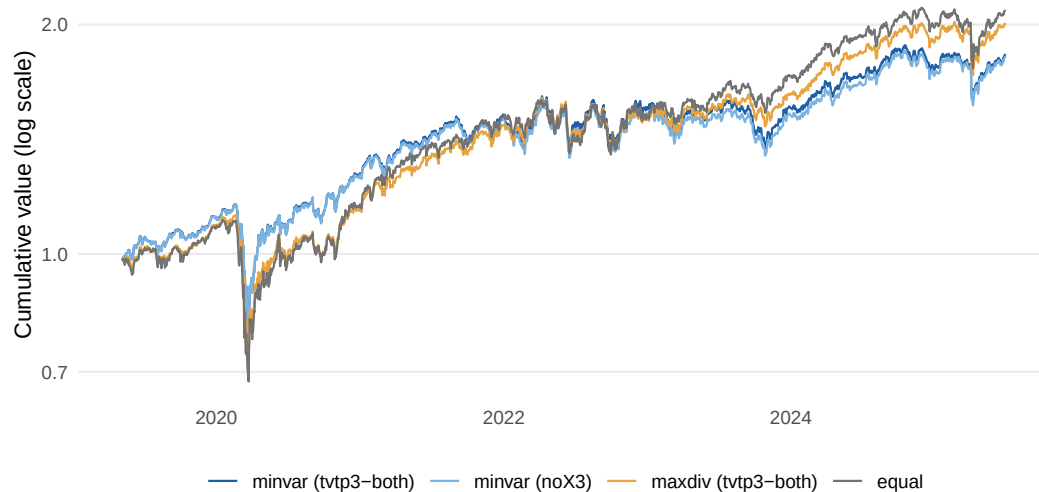


Figure 5: Cumulative value of the strategies on the out-of-sample window (log scale).

Both regime-based minimum-variance portfolios deliver a markedly lower realized volatility than the equal-weight benchmark on data none of their inputs have seen, at $K = 5$, where the full correlation matrix drives the weights, and the two variants are nearly indistinguishable, so the portfolio gain comes from the regime structure itself rather than from the covariates. The volatility reduction is bought with lower average returns: Sharpe ratios end up close to the benchmark's, slightly below it for the minimum-variance portfolios. Minimum variance achieves exactly its stated objective, no more.

5.6 Timing

```
timing <- data.frame(
  model = names(fits),
  npar = sapply(fits, function(f) f$npar),
  minutes = round(sapply(fits, function(f) attr(f, "minutes")), 1),
  row.names = NULL)
knitr::kable(timing, caption = sprintf(
  "Wall-clock estimation time per specification (four independent
  searches per fit on %d cores; T = %d training observations, K = %d). The C++
  filter evaluates one likelihood in about 2 ms at this sample size.",
  ctrl$cores, cut, ncol(z)))
```


Table 6: Wall-clock estimation time per specification (four independent searches per fit on 14 cores; $T = 3608$ training observations, $K = 5$). The C++ filter evaluates one likelihood in about 2 ms at this sample size.

model	npar	minutes
const	10	0.1
noX(N=2)	22	0.4
noX(N=3)	36	1.0
tvtp2-MCCC	24	0.6
tvtp2-VIX	24	0.6
tvtp2-both	26	0.7
tvtp3-MCCC	42	4.5
tvtp3-VIX	42	4.6
tvtp3-both	48	5.9

Estimation cost is driven by the number of parameters through the global stage's population size; the multi-start protocol runs its searches in parallel, so its wall-clock cost is close to that of a single search.

6 Conclusion

The **RSDC** package estimates regime-switching correlation models with constant, fixed-transition and covariate-driven transition dynamics, for any number of regimes and, since version 1.7-0, in cross-sectional dimensions where box-constrained search was previously infeasible, the global stage operates in a partial-correlation space where every candidate is a valid model (Joe, 2006). A shipped Monte Carlo study documents parameter recovery across all specifications and shows accuracy *improving* with the cross-section; a five-industry out-of-sample application favors three regimes on every criterion; within the three-regime class, in-sample criteria and likelihood-ratio tests support transitions driven by two nearly orthogonal indicators, market fear associated with entries into the high-correlation regime, climate attention predominantly with staying out of it, while the out-of-sample predictive likelihood ranks the fixed-transition variant marginally ahead, a reminder that in-sample transition fit need not convert into forecast gains. The regime-based minimum-variance portfolios attain the lowest out-of-sample volatility among the strategies considered, with or without covariates.

Some limitations should be kept in mind. Regime labels rest on the ascending-correlation ordering; all results are conditional on the univariate GARCH pre-filter; TVTP coefficients can reach their bounds in small samples when some transitions are rarely observed (the separation issue quantified in the validation study); and the empirical exercise is a software illustration on one asset universe, not a climate-finance study. Within those bounds, the package provides a reproducible pipeline from standardized residuals to regime-aware covariance forecasts and portfolio weights, driven by data shipped with the package.

Appendix: summary of the retained model

The full `summary()` of the retained TVTP($N = 3$) model, estimated on the training window, reports the softmax coefficients with observed-information standard errors, the implied transition matrix at the covariate means with delta-method standard errors, and per-regime diagnostics (stay probability, expected duration, ergodic probability). The coefficient naming follows `b[g<from>,c<softmax column>,x<covariate>]`, with x_1 the intercept, x_2 the MCCC and x_3 the log-VIX.

```
#> RSDC fit summary: method = "tvtp", N = 3, K = 5, T = 3608
#> logLik = -18941.097   npar = 48   AIC = 37978.19   BIC = 38275.36
#>
#> Coefficients:
#>           Estimate Std. Error z value Pr(>|z|)
#> b[g1,c1,x1]  2.831654   1.574766   1.798   0.07215 .
```

```

#> b[g1,c1,x2] 1.523738 1.124271 1.355 0.17532
#> b[g1,c1,x3] -2.148417 0.870985 -2.467 0.01364 *
#> b[g1,c2,x1] 3.381046 1.417505 2.385 0.01707 *
#> b[g1,c2,x2] 1.038269 1.126367 0.922 0.35664 .
#> b[g1,c2,x3] -1.189210 0.659733 -1.803 0.07146 .
#> b[g2,c1,x1] 0.490116 0.840901 0.583 0.56000
#> b[g2,c1,x2] 0.674542 0.320642 2.104 0.03540 *
#> b[g2,c1,x3] -1.278905 0.454219 -2.816 0.00487 **
#> b[g2,c2,x1] 3.146256 0.420711 7.478 7.52e-14 ***
#> b[g2,c2,x2] 0.692092 0.320008 2.163 0.03056 *
#> b[g2,c2,x3] -0.134730 0.305904 -0.440 0.65962
#> b[g3,c1,x1] -3.810217 0.906933 -4.201 2.65e-05 ***
#> b[g3,c1,x2] 0.377078 0.665168 0.567 0.57079
#> b[g3,c1,x3] 0.049939 0.480542 0.104 0.91723
#> b[g3,c2,x1] -2.057307 0.403054 -5.104 3.32e-07 ***
#> b[g3,c2,x2] -1.064365 0.508280 -2.094 0.03626 *
#> b[g3,c2,x3] -1.657270 0.515521 -3.215 0.00131 **
#> rho[g1,12] 0.400052 0.059606 6.712 1.93e-11 ***
#> rho[g1,13] 0.505599 0.090947 5.559 2.71e-08 ***
#> rho[g1,14] 0.306916 0.095930 3.199 0.00138 **
#> rho[g1,15] 0.210708 0.052876 3.985 6.75e-05 ***
#> rho[g1,23] 0.132076 0.075833 1.742 0.08157 .
#> rho[g1,24] 0.072515 0.060035 1.208 0.22709
#> rho[g1,25] 0.096280 0.059248 1.625 0.10415
#> rho[g1,34] 0.306405 0.093931 3.262 0.00111 **
#> rho[g1,35] 0.040424 0.050697 0.797 0.42524
#> rho[g1,45] 0.029735 0.055823 0.533 0.59426
#> rho[g2,12] 0.717074 0.029292 24.480 < 2e-16 ***
#> rho[g2,13] 0.857851 0.012326 69.596 < 2e-16 ***
#> rho[g2,14] 0.744952 0.022164 33.611 < 2e-16 ***
#> rho[g2,15] 0.608691 0.046421 13.112 < 2e-16 ***
#> rho[g2,23] 0.571862 0.041103 13.913 < 2e-16 ***
#> rho[g2,24] 0.498853 0.046105 10.820 < 2e-16 ***
#> rho[g2,25] 0.588250 0.043976 13.377 < 2e-16 ***
#> rho[g2,34] 0.749432 0.020168 37.160 < 2e-16 ***
#> rho[g2,35] 0.508456 0.055020 9.241 < 2e-16 ***
#> rho[g2,45] 0.533981 0.048090 11.104 < 2e-16 ***
#> rho[g3,12] 0.914371 0.008249 110.844 < 2e-16 ***
#> rho[g3,13] 0.937986 0.005229 179.370 < 2e-16 ***
#> rho[g3,14] 0.894696 0.010012 89.361 < 2e-16 ***
#> rho[g3,15] 0.866089 0.014660 59.079 < 2e-16 ***
#> rho[g3,23] 0.859352 0.013356 64.342 < 2e-16 ***
#> rho[g3,24] 0.829703 0.018445 44.982 < 2e-16 ***
#> rho[g3,25] 0.858007 0.016570 51.781 < 2e-16 ***
#> rho[g3,34] 0.877008 0.010544 83.174 < 2e-16 ***
#> rho[g3,35] 0.823629 0.016228 50.752 < 2e-16 ***
#> rho[g3,45] 0.859736 0.021214 40.526 < 2e-16 ***
#> ---
#> Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#>
#> Transition probabilities at the in-sample covariate means (delta-method SE):
#> entry estimate se
#> 1 p[1,1] 0.35828 0.12626
#> 2 p[1,2] 0.62061 0.11977
#> 3 p[1,3] 0.02111 0.03004
#> 4 p[2,1] 0.06308 0.03157
#> 5 p[2,2] 0.89829 0.02433
#> 6 p[2,3] 0.03864 0.01640
#> 7 p[3,1] 0.01926 0.01757
#> 8 p[3,2] 0.11113 0.04080
#> 9 p[3,3] 0.86961 0.03418
#>
#> Regime diagnostics at the in-sample covariate means (stay prob., expected duration, ergodic prob.):
#> regime stay_prob exp_duration ergodic_prob
#> 1 1 0.3583 1.558 0.07577
#> 2 2 0.8983 9.831 0.70350
#> 3 3 0.8696 7.669 0.22073

```

Computational details

Results were obtained with **RSDC** version 1.7.0, using **rugarch** for GJR-GARCH filtering, **DEoptim** for the global stage, and **quadprog** / **Rsolnp** for the portfolio optimizers. All

computations use the default estimation seed (123) and are reproducible from the packaged greenbrown, mccc and ff5ind data sets.

Acknowledgments

We gratefully acknowledge financial support from the Fonds de recherche AMF-GIRIF and the Natural Sciences and Engineering Research Council of Canada (grant RGPIN-2022-03767).

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David Ardia
CIRANO, GERAD, HEC Montréal
Department of Decision Science
Montréal, Canada
ORCID: [0000-0003-2823-782X](https://orcid.org/0000-0003-2823-782X)
david.ardia@hec.ca

Benjamin Seguin
HEC Montréal
Department of Decision Science
Montréal, Canada

Roosevelt Ymele Nguemo
HEC Montréal
Department of Decision Science
Montréal, Canada
ORCID: [0009-0003-9399-6256](https://orcid.org/0009-0003-9399-6256)